Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/07/2022
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	12/01/2015	**End Date:**	02/29/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	8	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G appended with that information per discussions with M. Sansoucie & L. Strutzenberg/MSFC (Ed., 6/21/19) NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/28/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0256 ; NNX16AB59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EDITOR's NOTE (June 2019): New grant number 80NSSC19K0256 awarded per NSSC information with period of performance of 3/1/2019-2/29/2024; this record for NNX16AB59G (POP of 12/1/2015-2/28/2019) appended with that information per discussions with M. Sansoucie & L. Strutzenberg/Marshall Space Flight Center (MSFC). Continuation of "Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events" ; grant NNX14AB74G. See that project for previous reports. Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS (International Space Station) Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights (PF) on selected alloys. Long Duration Microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS ; Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	• Graduated 4 PhD and 1 MS students. • Published 5 peer-reviewed articles; made 2 submissions to the 2021 Decadal Survey; & 7 presentations (see list at end of document). [Ed. Note. The two submissions/white papers to The National Academies of Sciences, Engineering, and Medicine Decadal Survey on Biological and Physical Sciences Research in Space 2023-2032 are listed in the Bibliography are listed as NASA Technical Documents and can be accessed through the NASA Technical Reports Server (NTRS).] • Selected as Chair of the International Space Station National Laboratory (ISS-NL) User Advisory Committee supporting Center for the Advancement of Science in Space (CASIS) Ops. • Selected as Co-chair of the National Academies Committee on Biological and Physical Sciences in Space to oversee the 2022 NASA Decadal Study. • Conducted flight testing during International Space Station (ISS) Expedition 66 using the European Space Agency (ESA) ISS Electromagnetic Levitator (EML) facility processing Titanium Aluminum (#66) and high entropy alloy FeCrNiCo (#83). • Attended the 37th Annual American Society for Gravitational and Space Research (ASGSR-37) 2021 Annual Conference, November 3-6 as organizer, and recipient of Fellow Award. • Featured speaking engagements at: International Space Station Research and Development (ISSR&D) Conference Plenary Panel; NASA Headquarters (HQ) Kibo Utilization Symposium; Japan Microgravity Application Society Conference (JASMAC-33); and the United Nations Office for Outer Space Affairs Webinar #5 – Physical Sciences: Materials Science. • Conducted flight planning for Batch 3 lead on samples FeCrNi 16-12 (#84) and FeCrNi 12-16 (#86) with ESA ThermoProp, and CoSi eutectic (#52), in collaboration with the ESA Liquid Phase Separation in Metallic Alloys (Li-PHase) project.
Bibliography Type:	Description: (Last Updated: 05/05/2026)

Articles in Peer-reviewed Journals	Wunderlich RK, Mohr M, Dong Y, Hecht U, Matson DM, Hyers R, Bracker G, Lee J, Schneider S, Xiao X, Fecht HJ. "Thermophysical properties of the TiAl-2Cr-2Nb alloy in the liquid phase measured with an electromagnetic levitation device on board the International Space Station ISS-EML." Int. J. Mater. Res. 2021;112(10):770-781. https://doi.org/10.1515/ijmr-2021-8266 , Oct-2021
Articles in Peer-reviewed Journals	Xiao X, Brillo J, Lee J, Hyers RW, Matson DM. "Impact of convection on the damping of an oscillating droplet during viscosity measurement using the ISS-EML facility." npj Microgravity. 2021;7:(36):1-7. https://doi.org/10.1038/s41526-021-00166-4 , Oct-2021
Articles in Peer-reviewed Journals	Jeon S, Liu X, Azersky C, Ren J, Zhang S, Chen W, Hyers RW, Costa K, Kolbe M, Matson DM. "Particle size effects on dislocation density, microstructure, and phase transformation for high-entropy alloy powders." Materialia. 2021 Jul 13;18:101161. https://doi.org/10.1016/j.mtla.2021.101161 , Jul-2021
Articles in Peer-reviewed Journals	Matson DM, Liu X, Rodriguez JE, Jeon S, Shuleshova O. "Retained free energy with enhanced nucleation during electrostatic levitation of undercooled Fe-Co alloys." Crystals. 2021 Jun 24;11(7):730. https://doi.org/10.3390/cryst11070730 , Jun-2021
Articles in Peer-reviewed Journals	Baker EB, Jeon S, Shuleshova O, Kaban I, Wang Y, Gao J, Kolbe M, Sansoucie MP, Matson DM. "Dendrite remelting during rapid solidification of undercooled CoSi-CoSi2 eutectic alloys quantified by in situ synchrotron X-ray diffraction." Scr. Mater. 2021 Mar 15;194:113645. https://doi.org/10.1016/j.scriptamat.2020.113645 , Mar-2021
Awards	Matson DM. "American Society for Gravitational and Space Research (ASGSR) Fellow, November 2021." Nov-2021
Dissertations and Theses	Nawer J. "Thermophysical property measurement using containerless techniques in microgravity and terrestrial environment." Dissertation, Tufts University, January 2022. , Jan-2022
Dissertations and Theses	Baker E. "Mechanisms for metastable phase transformations in CoSi and FeCo alloys." Dissertation, Tufts University, August 2021. , Aug-2021
Dissertations and Theses	Bracker G. Coadvised with R.W. Hyers. "Solidification experiments and magnetohydrodynamic models in electromagnetic levitation." Dissertation, University of Massachusetts-Amherst, November 2021. , Nov-2021
Dissertations and Theses	Leitner T. Coadvised with G. Pottlacher. "Thermophysical property measurement of industrial metals and alloys using electromagnetic levitation." Dissertation, Graz University of Technology, May 2021. , May-2021
NASA Technical Documents	Narayanan R, Banish M, Derby J, Hages C, Ishikawa T, Kelton K, Matson D, Matsumoto S, McCormack S, Paradis P-F, Rodocea A, Raush J, Sansoucie M, Savin K, Singh NB, Volk J, Weber R, Wilke S, Ziegler K. "Thermophysical and Thermochemical Properties – The Basis for All Advanced Materials Research." White Paper for the Decadal Survey on Life and Physical Sciences Research in Space 2023-3032, National Academies of Sciences, Engineering, and Medicine. Huntsville, Alabama : NASA Marshall Space Flight Center, 2021. NASA/Document ID# 20210023248. https://ntrs.nasa.gov/citations/20210023248 , Oct-2021
NASA Technical Documents	Hanson I, Biggs R, Fody J, Glaessgen E, Kinsey B, Matson D, Narayanan R, Radocea A, Richter B, Roth JT, Sansoucie M, Stott J, Strutzenberg L, Voorhees P, Wegst U. "Research Campaign: The Sciences of Space Manufacturing. " White Paper for the Decadal Survey on Biological and Physical Sciences Research in Space 2023-3032, National Academies of Sciences, Engineering, and Medicine. Huntsville, Alabama : NASA Marshall Space Flight Center, 2021. NASA/Document ID# 20210025971. https://ntrs.nasa.gov/citations/20210025971 , Dec-2021